

Counters

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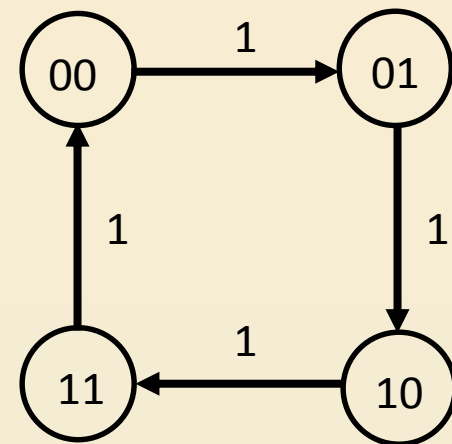
Definition:

- **Counter** : A sequential circuit that goes through prescribed sequence of states upon the application of clock pulse is called a counter.
- The input pulses are called **count pulses**, may be clock pulses or they may originate from an external source or occur at prescribed intervals or at random.
- In a counter the sequence of states may follow binary count or any other sequence.
- Counters are found in almost all equipment containing digital system

Binary Counter :

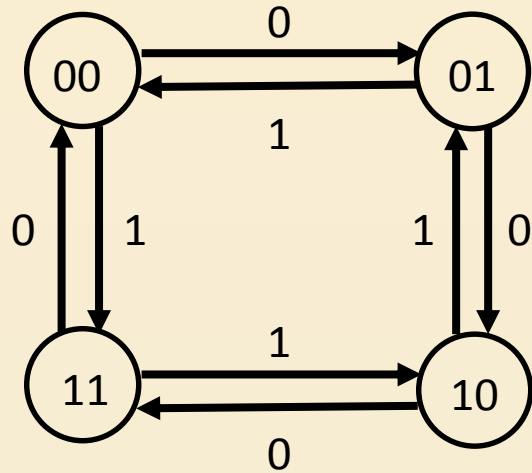
- **Binary Counter** : A counter that follows the binary sequence is called a Binary counter.
- An n-bit Binary Counter consists of n flipflops and can count in binary from **0 to 2^n-1** .
- For eg. Binary counter for two digits is as follows:

Present State		Next State	
A	B	A	B
0	0	0	1
0	1	1	0
1	0	1	1
1	1	0	0



Example of binary counter

- Here's the another example for binary counter and its state table:



Present State		Inputs X	Next State	
Q ₁	Q ₀		Q ₁	Q ₀
0	0	0	0	1
0	0	1	1	1
0	1	0	1	0
0	1	1	0	0
1	0	0	1	1
1	0	1	0	1
1	1	0	0	0
1	1	1	1	0

Applications of Counters

- Some of the applications of counters in a sequential circuits are as follows:
 - To count the number of occurrences
 - Generating Timing sequences
 - Count up or down
 - Increment or decrement count
 - Sequence events
 - Divide frequency
 - Address memory
 - As temporary memory

Two principal categories

- *Counters are divided in two categories, these are:*
 - ***Asynchronous (Ripple) Counters*** - the first flip-flop is clocked by the external clock pulse, and then each successive flip-flop is clocked by the Q or Q' output of the previous flip-flop.
 - ***Synchronous Counters*** - all memory elements are simultaneously triggered by the same clock.

Few other categories of counters:

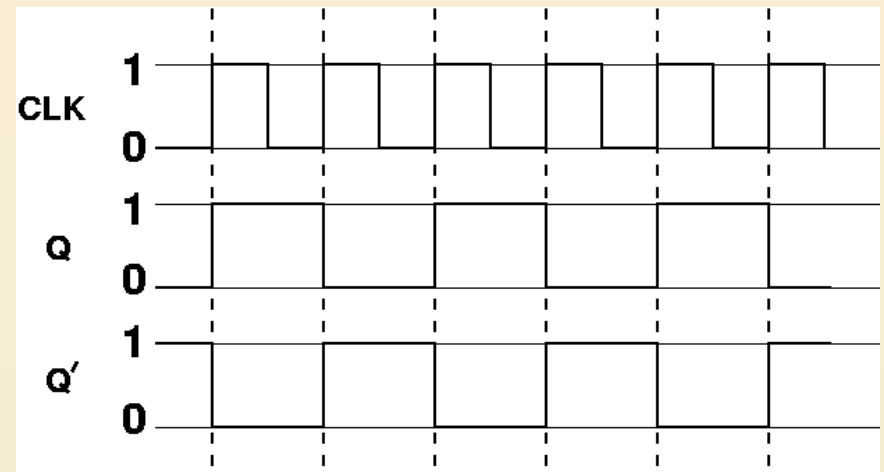
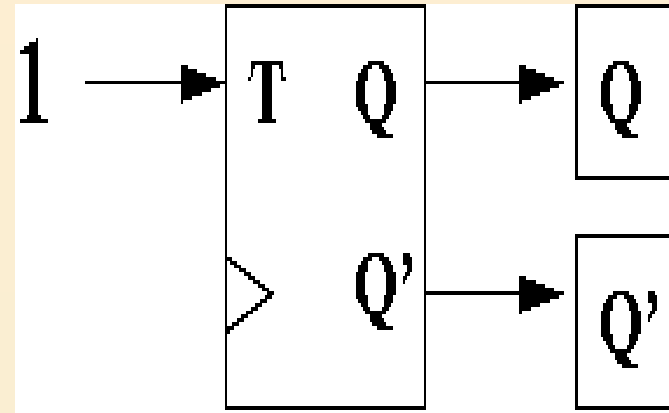
- Apart from synchronous and asynchronous counters which are the major ones the other types of counters are as follows:
 - *Ring counter*
 - *Johnson counter*
 - *Decade counter*
 - *Up–down counter*

Asynchronous Counters:

- Here flipflop output transition serves as a source for triggering other flipflops.
- This means that the clockpulse is provided to a single flipflop
- The change of state of a given flipflop is dependent on the states of other flipflops.
- In other words the flipflops are not triggered by simultaneous clock pulses but the transitions in other flipflops.

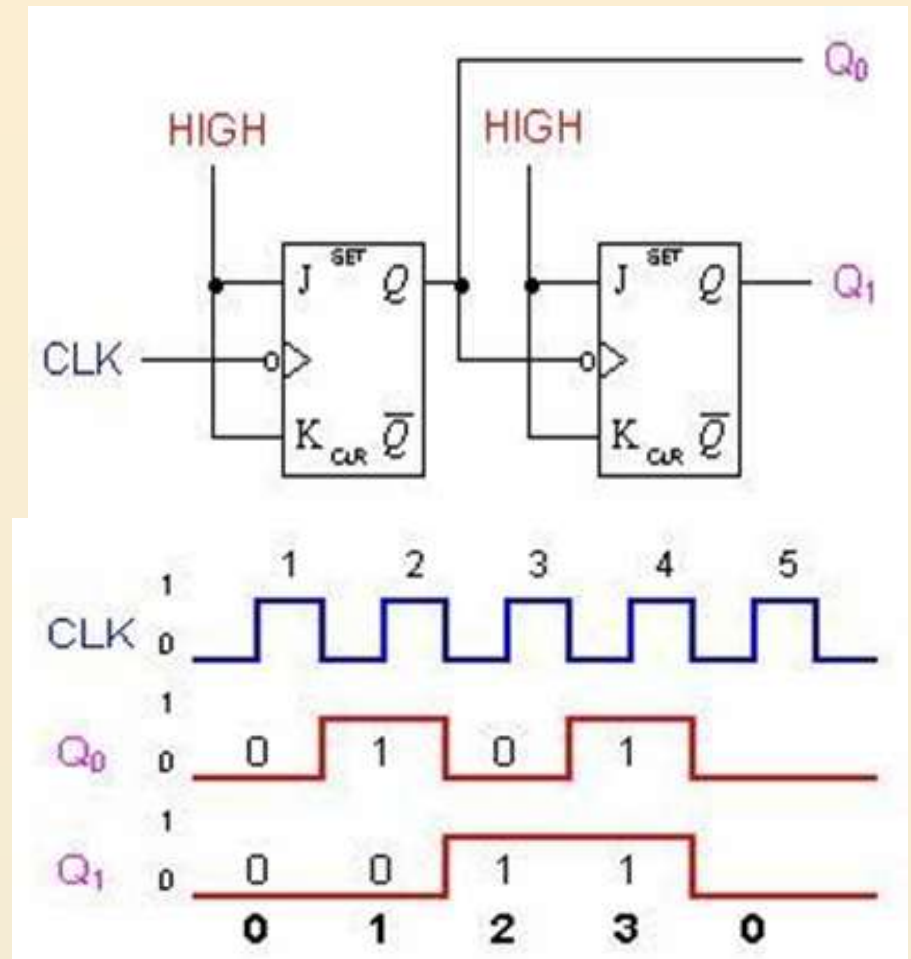
Asynchronous Counters

- This counter is called *asynchronous* because not all flip flops have the same clock.
- Look at the waveform of the output, **Q**, in the timing diagram. It resembles a clock as well.
- This provides a clock that runs twice as slow. If we feed the clock into a T flip flop, where T is hardwired to 1. The output will be a clock whose period is twice as long.



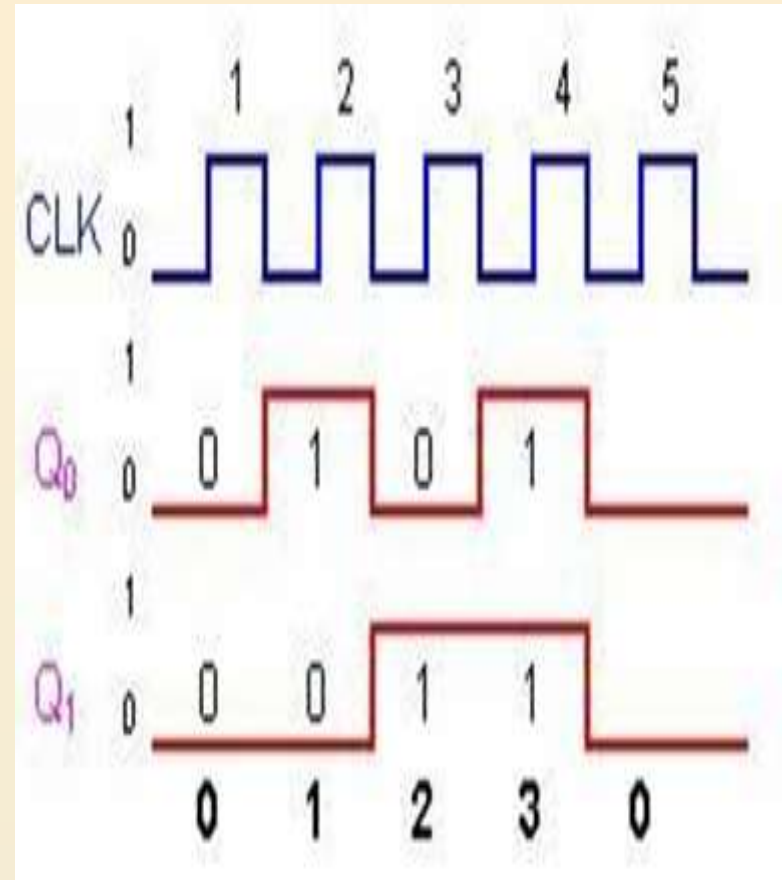
Two-bit asynchronous counter

- The external clock is connected to the clock input of the first flip-flop only.
- So, it changes state at the *negative edge* of each clock pulse, but the next flipflop changes only when triggered by the negative edge of the Q output of the first one.



Two-bit asynchronous counter

- Because of the inherent propagation delay through a flip-flop, the transition of the input clock pulse and a transition of the Q output of FF0 **can never occur at exactly the same time**.
- Therefore, the flip-flops cannot be triggered simultaneously, producing an asynchronous operation.
- Eg : As shown, there is some small delay between the CLK, Q0 and Q1 transitions.

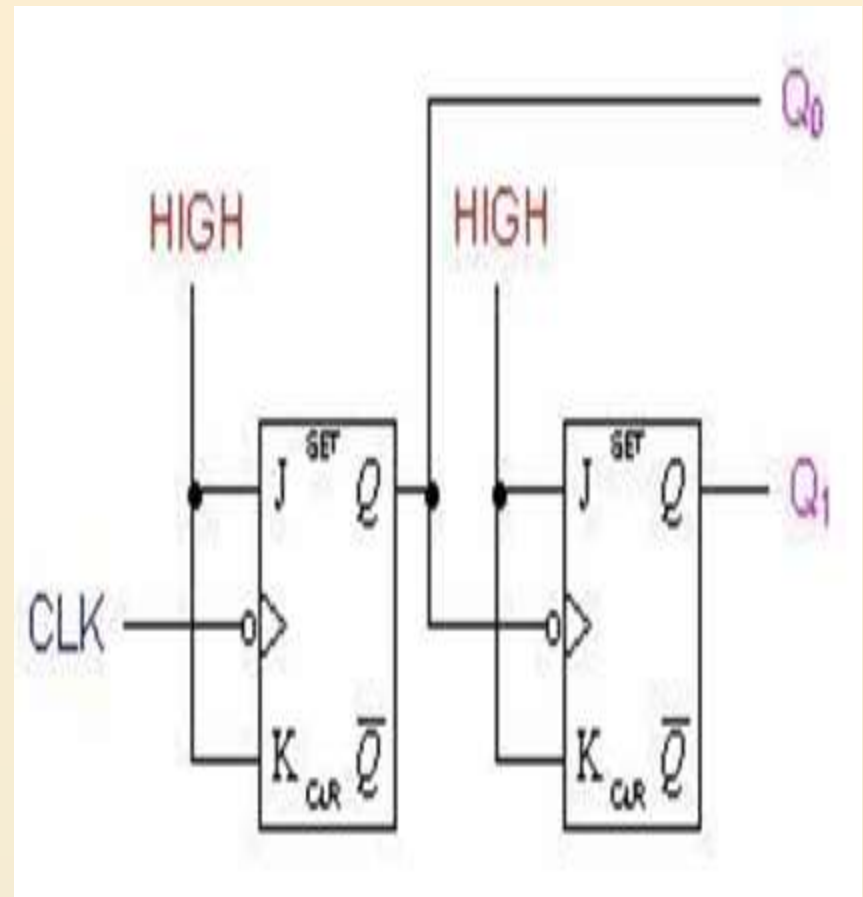


Two-bit asynchronous counter

- Usually, all **the CLEAR inputs** are connected together, so that a single pulse can clear all the flip-flops before counting starts.
- The 2-bit ripple counter circuit above has four different states, each one corresponding to a count value.
- Similarly, a counter with n flip-flops can have 2^N **states**.
- The number of states in a counter is known as its mod (modulo) number.

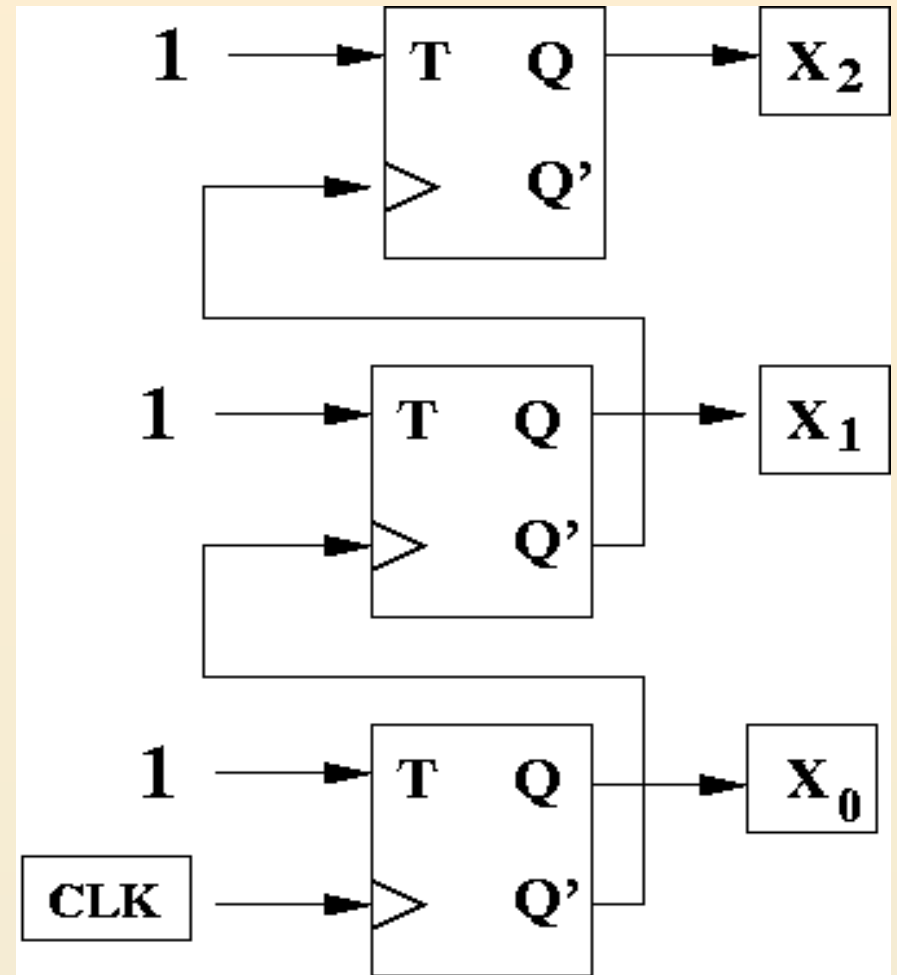
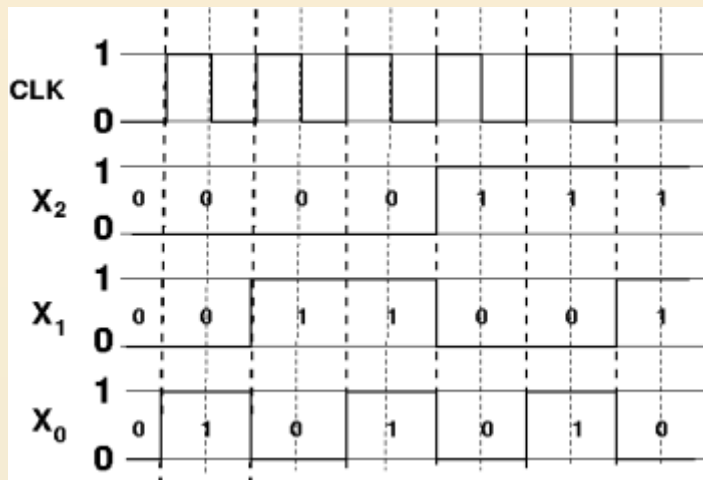
Two-bit asynchronous counter

- Thus a 2-bit counter is a mod-4 counter.
- This is because the most significant flip-flop produces one pulse for every n pulses at the clock input of the least significant flip-flop .



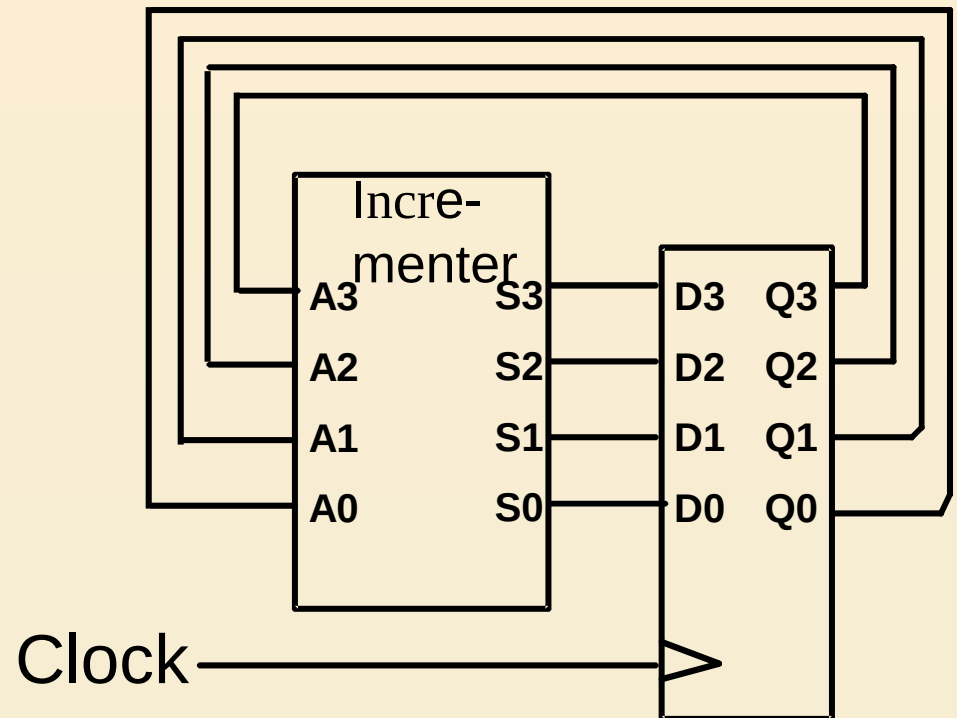
3 bit asynchronous “ripple” counter using T flip flops

- This is called as a *ripple counter* due to the way the FFs respond one after another in a kind of rippling effect.



Synchronous Counters

- To eliminate the "ripple" effects, use a common clock for each flip-flop and a combinational circuit to generate the next state.
- For an up-counter, use an incrementer =>

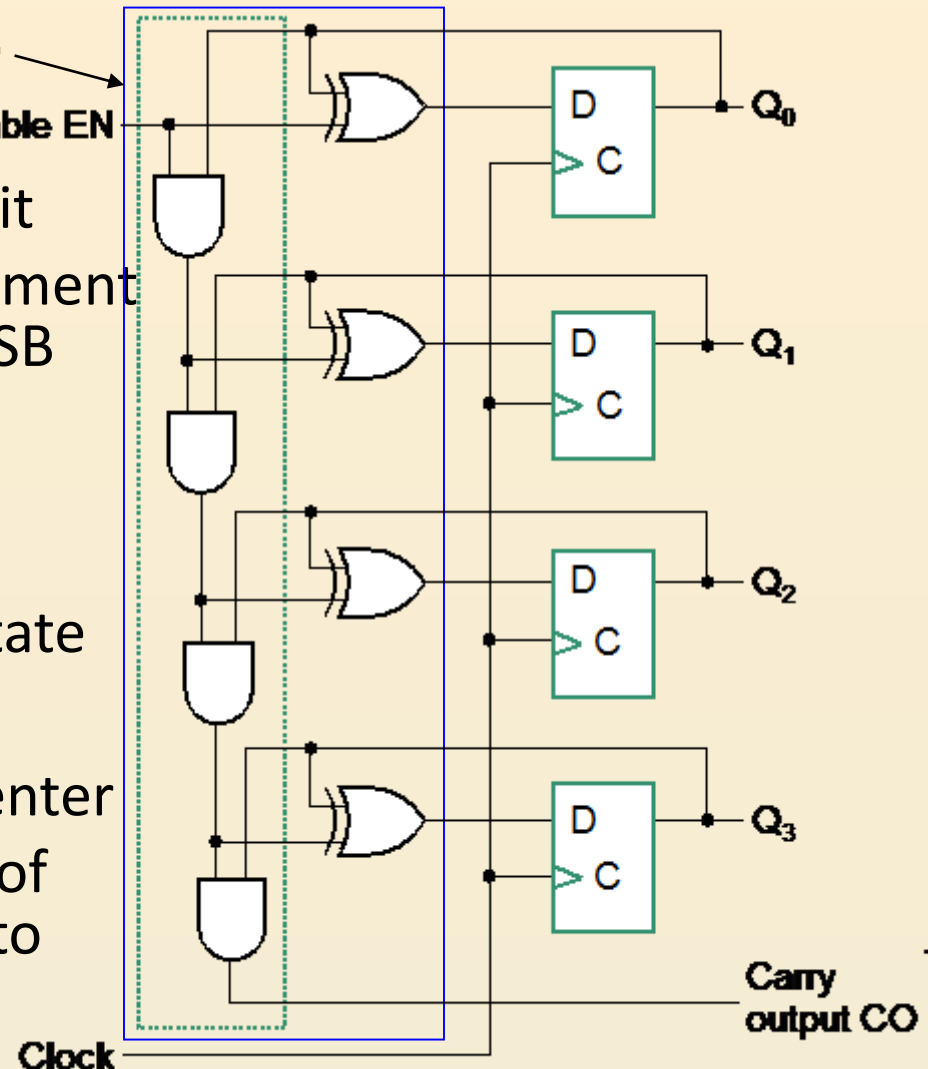


Synchronous Counters

- To eliminate the "ripple" effects, use a common clock for each flip-flop and a combinational circuit to generate the next state.
- Hence the counters in which all the flipflops are provided with a clock pulse simultaneously are called the **Synchronous counters**.
- Synchronous counters may be of the following types
 - Up counter
 - Down counter

Synchronous Counters (continued)

- Internal details => Incrementer
- Internal Logic
 - XOR complements each bit
 - AND chain causes complement of a bit if all bits toward LSB from it equal 1
- Count Enable
 - Forces all outputs of AND chain to 0 to “hold” the state
- Carry Out
 - Added as part of incrementer
 - Connect to Count Enable of additional 4-bit counters to form larger counters



(a) Logic Diagram-Serial Gating

Design Example: Synchronous BCD

- Use the sequential logic model to design a synchronous BCD counter with D flip-flops
- State Table =>
- Input combinations 1010 through 1111 are don't cares

Current State				Next State			
Q8	Q4	Q2	Q1	Q8	Q4	Q2	Q1
0	0	0	0	0	0	0	1
0	0	0	1	0	0	1	0
0	0	1	0	0	0	1	1
0	0	1	1	0	1	0	0
0	1	0	0	0	1	0	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	1	1
0	1	1	1	1	0	0	0
1	0	0	0	1	0	0	1
1	0	0	1	0	0	0	0

Synchronous BCD (continued)

- Use K-Maps to two-level optimize the next state equations and manipulate into forms containing XOR gates:

$$D1 = Q1'$$

$$D2 = Q2 \oplus Q1Q8'$$

$$D4 = Q4 + Q1Q2$$

$$D8 = Q8 + (Q1Q8 + Q1Q2Q4)$$

- $Y = Q1Q8$

- The logic diagram can be drawn from these equations
 - An asynchronous or synchronous reset should be added

Synchronous BCD (continued)

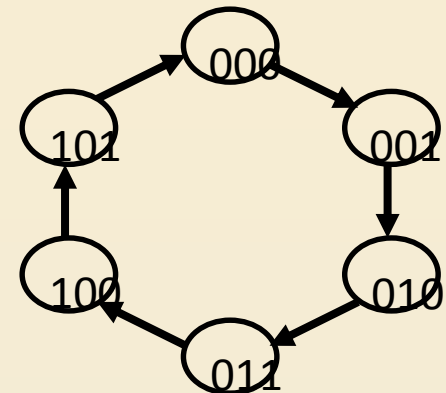
- There are chances that counter is perturbed by a power disturbance or other interference and it enters a **state other than 0000 through 1001**.
- Find the actual values of the six next states for the don't care combinations from the equations
- Find the overall state diagram to assess behavior for the don't care states (states in decimal)

Present State	Next State
Q8 Q4 Q2 Q1	Q8 Q4 Q2 Q1
1 0 1 0	1 0 1 1
1 0 1 1	0 1 1 0
1 1 0 0	1 1 0 1
1 1 0 1	0 1 0 0
1 1 1 0	1 1 1 1
1 1 1 1	0 0 1 0

Unused states

- The examples shown so far have all had 2^n states, and used n flip-flops. But sometimes you may have unused, leftover states.
- For example, here is a state table and diagram for a counter that repeatedly counts from 0 (000) to 5 (101).
- But there are also the unused states in the state table.

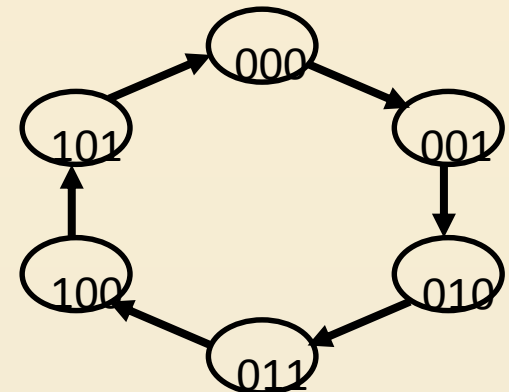
Present State			Next State		
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	0	0	0
1	1	0	?	?	?
1	1	1	?	?	?



Unused states can be don't cares...

- To get the *simplest* possible circuit, one can fill in don't cares for the next states. This will also result in don't cares for the flip-flop inputs, which can simplify the hardware.
- If the circuit somehow ends up in one of the unused states (110 or 111), its behavior will depend on exactly what the don't cares were filled in with.

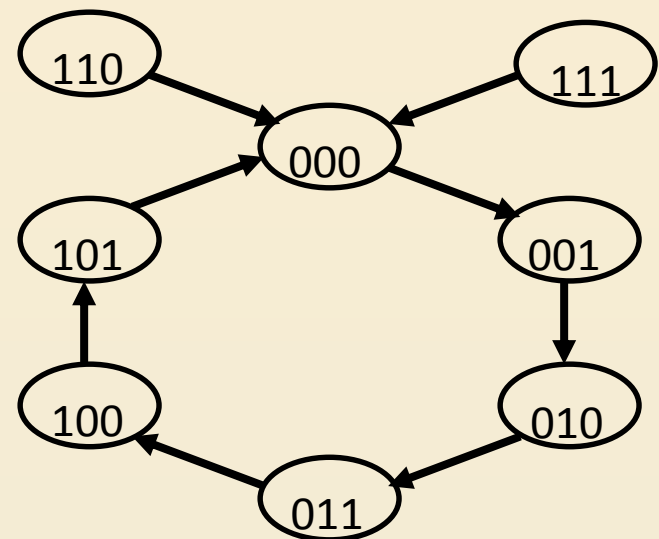
Present State			Next State		
Q ₂	Q ₁	Q ₀	Q ₂	Q ₁	Q ₀
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	0	0	0
1	1	0	x	x	x
1	1	1	x	x	x



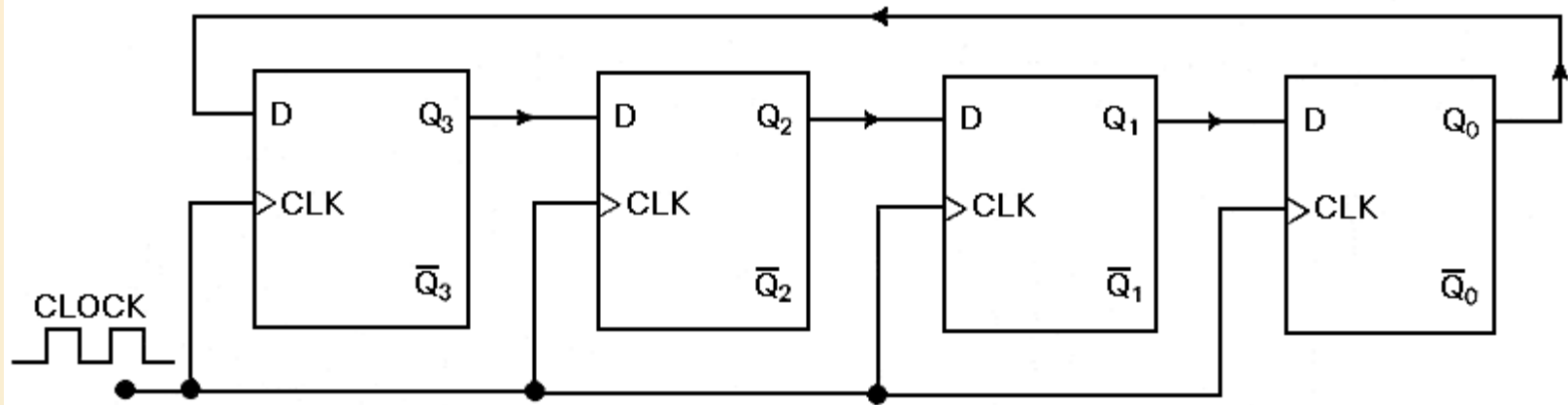
Other possibility

- To get the *safest* possible circuit, you can explicitly fill in next states for the unused states 110 and 111.
- This guarantees that even if the circuit somehow enters an unused state, it will eventually end up in a valid state.
- This is called a **self-starting counter**.

Present State			Next State		
Q ₂	Q ₁	Q ₀	Q ₂	Q ₁	Q ₀
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	0	0	0
1	1	0	0	0	0
1	1	1	0	0	0

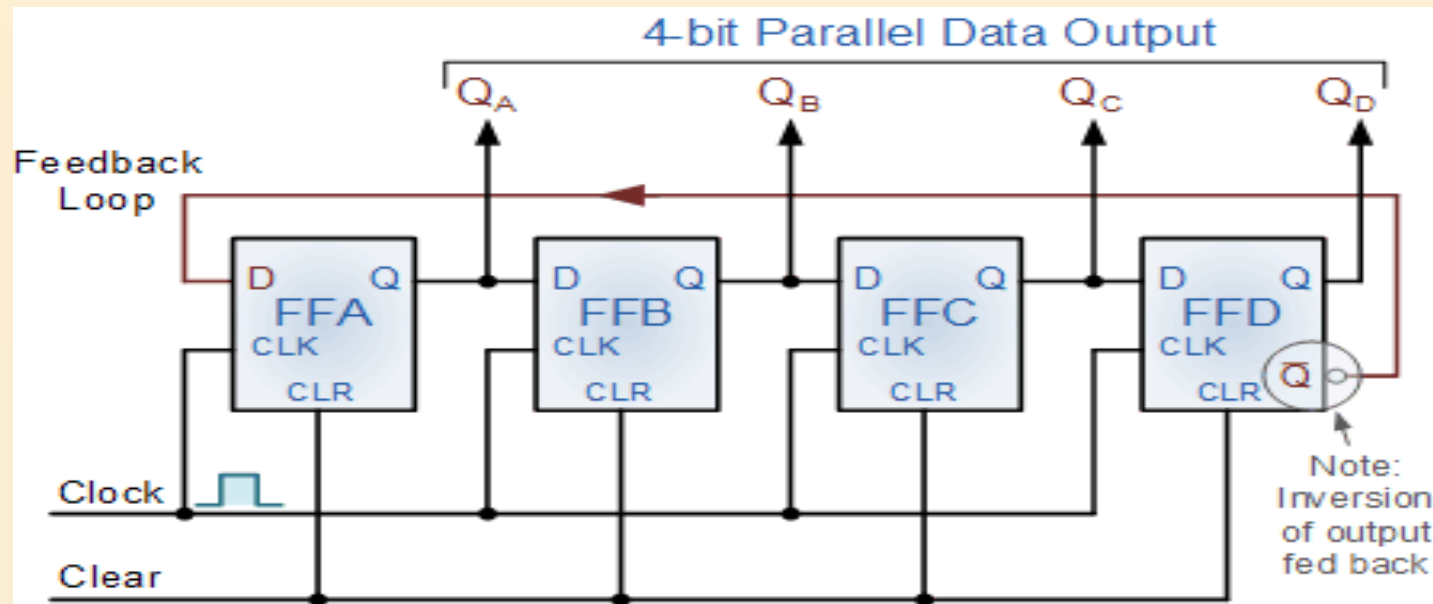


Ring Counter



Ring counters are implemented as shown where output of a flipflop is given as an input to the next flipflop except the last one. Here the normal output of the last flipflop is connected back to the input of the first one. Hence it is called Ring counter.

Johnson Counter



The Johnson counter, also known as the *twisted-ring counter*, is exactly the same as the ring counter except that the inverted output of the last flip-flop is connected to the input of the first flip-flop.

If it starts from 000, 100, 110, 111, 011 and 001, and the sequence is repeated so long as there is input pulse.

Thank You!

